

Project Title

Marketing Campaign Performance & Customer Conversion Analysis

Problem Statement

Businesses run multiple marketing campaigns but struggle to understand:

- Which campaigns generate the highest conversions
- Which customer segments are most valuable
- Which sales channels perform best

This project aims to analyze customer and campaign data to identify patterns, improve targeting, and increase ROI.

Dataset Overview

Source: Kaggle (Customer Personality Analysis Dataset)

Context: The dataset represents customer behavior data from a retail company. It is designed to help analyze customer personality traits, purchasing behavior, and response to marketing campaigns in order to improve targeting and product strategies.

By understanding different customer segments, businesses can optimize marketing efforts instead of targeting all customers uniformly.

Data Structure: The dataset is divided into four main categories:

1. Customer Demographics

- ID – Unique customer identifier
- Year_Birth – Year of birth (used to calculate age)
- Education – Education level
- Marital_Status – Marital status
- Income – Annual household income
- Kidhome – Number of children in household
- Teenhome – Number of teenagers in household
- Dt_Customer – Date of customer enrollment

2. Product Spending Behavior

- MntWines – Spending on wine
- MntFruits – Spending on fruits
- MntMeatProducts – Spending on meat
- MntFishProducts – Spending on fish
- MntSweetProducts – Spending on sweets
- MntGoldProds – Spending on premium products

3. Marketing Campaign Response

- AcceptedCmp1 – Campaign 1 response
- AcceptedCmp2 – Campaign 2 response
- AcceptedCmp3 – Campaign 3 response
- AcceptedCmp4 – Campaign 4 response
- AcceptedCmp5 – Campaign 5 response
- Response – Customer response to the most recent marketing campaign

4. Sales Channel Behavior

- NumDealsPurchases – Purchases made with discounts
- NumWebPurchases – Online purchases
- NumCatalogPurchases – Catalog purchases
- NumStorePurchases – In-store purchases
- NumWebVisitsMonth – Website visits per month

5. Customer Feedback

- Recency – Days since last purchase
- Complain – Complaint flag (1/0)
- Response – response

6. Operational Fields (Not used in analysis)

- Z_CostContact – Internal cost metric
- Z_Revenue – Internal revenue metric

4. Tools Used

- Microsoft Excel (data cleaning & analysis)
- Excel (visualization)
- Google Docs (documentation)

5. 🧹 Data Cleaning Process

Data cleaning was performed using Microsoft Excel to ensure data consistency and analysis readiness.

Data Preparation

- Created backup dataset
- Renamed sheets:
 - RAW_DATA
 - CLEANED_DATA
- All transformations performed on cleaned dataset

Removed Irrelevant Columns

- Z_CostContact
- Z_Revenue

Duplicate Check

Formula used:

```
=IF(COUNTIF($A$2:$A$2241, A2) > 1, "Duplicate", "Unique")
```

Result: No duplicates found

Missing Value Treatment

- Income → Median imputation
- Marital_Status → “Unknown”
- Education → “Unknown”

Categorical Data Cleaning (Feature Engineering)

Marital Status Standardization

```
=SWITCH(LOWER(TRIM(D2)),  
"married","Married",  
"together","In Relationship",  
"single","Single",  
"alone","Single",  
"divorced","Divorced",  
"widow","Widowed",  
"yolo","Unknown",  
"Unknown")
```

Education Standardization

```
=SWITCH(LOWER(TRIM(E2)),  
"graduation","Graduate",  
"phd","PhD",  
"master","Master",  
"2n cycle","Master",  
"basic","High School",  
"Unknown")
```

Date Processing

- Converted inconsistent date formats into proper date type
- Standardized Dt_Customer field
- Extracted:
 - Enrollment Year
 - Customer Tenure (Days)

```
=IF(ISNUMBER(S2), S2, DATE(RIGHT(S2,4), MID(S2,4,2), LEFT(S2,2)))
```

Feature Engineering

Age

```
=YEAR(TODAY()) - Year_Birth
```

Total Spending

=SUM(MntWines:MntGoldProds)

Total Purchases

=SUM(NumDealsPurchases:NumStorePurchases)

Total Campaign Responses

=SUM(AcceptedCmp1:AcceptedCmp5)

Customer Value Score

=Total Spend + Total_Purchases + (Total_Campaigns_Accepted*100)

Final Outcome of Cleaning

- Data inconsistencies resolved
- Missing values handled
- Categorical variables standardized
- New analytical features created

Dataset is now clean, structured, and ready for analysis.

6. Analysis Questions & Key Insights

The following business questions were analyzed:

1. Which customer segments generate the highest revenue?
 - Revenue by Income Segment:
High-income customers contribute the highest revenue, followed closely by mid-income customers, while low-income customers contribute significantly less. This indicates a strong positive relationship between income level and spending behavior.
 - Revenue by Age Segment:
Senior customers generate the highest total revenue, making them the most valuable age segment.
 - Customer Distribution by Age:
Senior customers form the largest group, which contributes to their higher total revenue.

- Purchase Behavior by Age:
Senior customers make the highest number of purchases, showing consistent buying behavior and contributing significantly to revenue.
2. Which sales channels perform best (Web vs Store vs Catalog)?
Store purchases dominate, indicating strong offline buying behavior. Web is the second most used channel, while catalog and deal-based purchases are less significant.
 3. Which marketing campaigns have the highest acceptance rate?
Campaign 4 has the highest acceptance rate, followed by Campaigns 3 and 5. Campaign 2 performs the weakest, indicating low effectiveness.
 4. What is the relationship between income and spending?
Higher-income segments consistently drive greater spending, suggesting strong potential for targeted premium marketing strategies.
 5. Who are the highest-value customers?
Top 10 customers show:
 - High customer value scores (~2400–2650)
 - Long tenure (4000+ days)
 - High engagement and frequent purchases
-

7. Key Visualizations

- Revenue by Age Group
 - Campaign Success Rate
 - Channel Performance Comparison
 - Income vs Spending Top Customer Segments
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8. Key Insights

- Customers aged 55+ show the highest spending behavior
 - Campaign 4 has the highest conversion rate
 - Store purchases generate higher average revenue than web purchases
 - High-income customers respond more positively to campaigns
 - A small segment of customers contributes to most revenue
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9. Business Recommendations

- Focus marketing campaigns on high-value customer segments
 - Improve underperforming campaigns or reallocate budget
 - Strengthen high-performing sales channels
 - Personalize offers for high-income customers
 - Optimize targeting strategy to improve conversion rate
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10. Conclusion

This analysis helped identify key customer segments, evaluate campaign effectiveness, and uncover opportunities to improve marketing ROI. The findings can be used to optimize future campaigns and increase revenue efficiency.

11. Appendix (Optional)

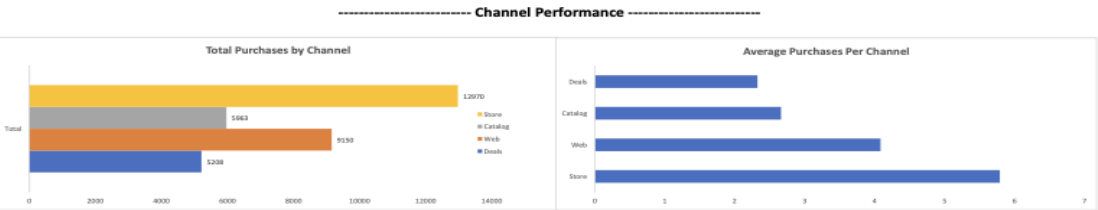
- Link to dataset
<https://www.kaggle.com/datasets/rodsaldanha/arketing-campaign>
- Dashboard screenshots

Marketing Campaign Performance Dashboard				
Customers	Total Revenue	Average Spend	Purchases	Conv Rate
2240	1356988	605.80	33291	15%

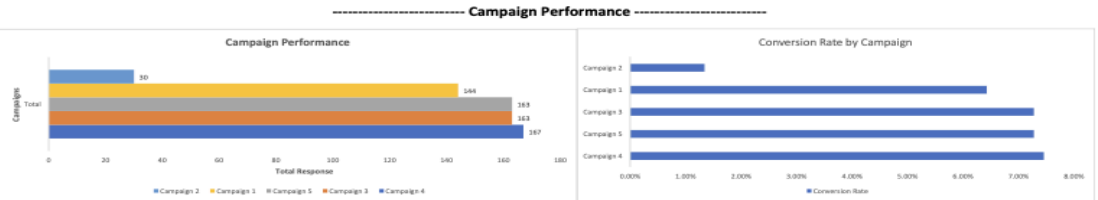
Insight: High-income & senior customers drive revenue. Store channel dominates. Campaign 4 has highest conversion.



High-income customers contribute the most revenue. Senior customers contribute more total revenue due to volume. Young customers spend more per transaction on average.

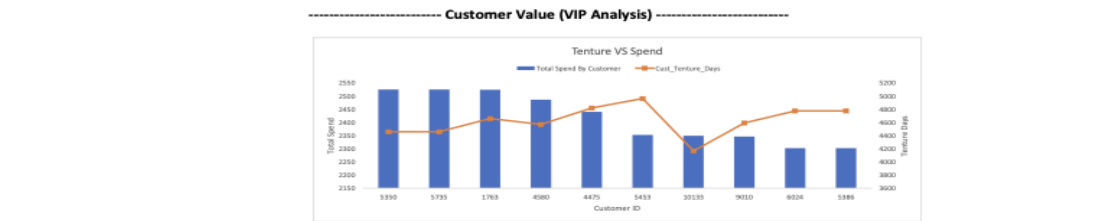


Store channel dominates total purchases, indicating strong offline purchasing behavior. Web channel is the second strongest, showing significant digital adoption, while catalog and deal-based purchases play a secondary role in the customer journey.

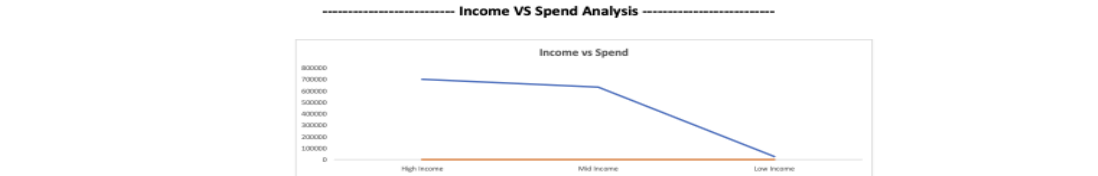


Campaign 4 has the highest acceptance rate (167 acceptances), making it the most successful campaign in terms of customer response. Campaigns 3 and 5 also perform strongly, with similar high acceptance levels (163 each), showing consistent engagement across these campaigns.

Campaign 1 shows moderate performance (144 acceptances), while Campaign 2 has the lowest acceptance (30), indicating weak customer response and low campaign effectiveness.



High-value customers show long tenure and consistent engagement



Higher income leads to higher spending.

----- Final Insights and Recommendations -----

****Key Insights****

- High-income customers contribute the majority of total revenue
- Senior customers represent the most valuable segment
- In-store purchases are the dominant sales channel
- Campaign 4 demonstrates the highest conversion performance

****Business Recommendations****

- Focus marketing efforts on high-income customers with premium targeting strategies
- Implement retention programs for senior customers to maximize lifetime value
- Optimize or redesign underperforming campaigns (Campaign 1 & 2)
- Strengthen both in-store and online channels to improve overall customer experience